TryHackMe - Windows Fundamentals 2

Platform: TryHackMe
Room link: Windows Fundamentals 2

1. Introduction

Windows includes several built-in security features designed to protect computers against malware, unauthorized access, and data loss.

This room explores important Windows security and maintenance features, including Windows Update, Microsoft Defender Antivirus, Windows Firewall, App & Browser Control, BitLocker, and Volume Shadow Copy.

Understanding these features is useful for system administration, cybersecurity investigations, and Windows hardening.

2. Learning Objectives

By completing this room, you should understand:

- How Windows Update works.
- How Windows Security protects a computer.
- The purpose of Microsoft Defender Antivirus.
- How Windows Firewall protects network connections.
- App & Browser Control and SmartScreen.
- Device security and hardware-based protections.
- BitLocker drive encryption.
- Volume Shadow Copy and file recovery.

3. Tools and Environment

The room primarily uses the Windows graphical interface.

Useful tools include:

Tool	Purpose
Windows Update	Install operating system updates
Windows Security	Manage security features
Microsoft Defender Antivirus	Detect and block malware
Windows Firewall	Control network traffic
BitLocker	Encrypt storage drives
PowerShell	Inspect system configuration
Disk Management	View storage volumes
File Explorer	Browse files and folders

4. Task-wise Walkthrough

Task 1 - Introduction

Windows provides multiple layers of security rather than relying on a single protection mechanism.

These layers include:

- Operating system updates.
- Antivirus and threat protection.
- Network firewall protection.
- Application and browser protection.
- Hardware and device security.
- Storage encryption.
- Backup and recovery mechanisms.

A secure Windows environment requires these features to work together.

Task 2 - Windows Update

Windows Update is Microsoft's service for delivering operating system updates.

Updates can include:

- Security patches.
- Bug fixes.
- Performance improvements.
- Driver updates.
- Feature updates.

To access Windows Update:

1. Open Settings.
2. Navigate to Update & Security.
3. Select Windows Update.
4. Review the update status and available options.

You can also open the Windows Update settings page using Run:

ms-settings:windowsupdate

Why updates matter

Unpatched systems may contain known vulnerabilities that attackers can exploit.

For example, an outdated operating system may be vulnerable to publicly documented security flaws.

Keeping Windows updated helps reduce this risk.

Practical exercise: Explore Windows Update and identify the available update settings. Do not interrupt an update that is already installing.

Task 3 - Windows Security

Windows Security is the central interface for managing several built-in security features.

It provides access to security areas such as:

- Virus & threat protection.
- Account protection.
- Firewall & network protection.
- App & browser control.
- Device security.
- Device performance and health.

You can open Windows Security from the Start Menu by searching for:
Windows Security

Its purpose is to make security settings and protection status accessible from one location.

Task 4 - Virus & Threat Protection

Microsoft Defender Antivirus is built into modern Windows versions.

It helps detect, prevent, and remove malicious software.

Common threats include:

- Viruses.
- Trojans.
- Ransomware.
- Spyware.
- Worms.
- Potentially unwanted applications.

Important protection features include:

- **Real-time protection**: Monitors files and applications as they are accessed or executed.
- **Cloud-delivered protection**: Uses Microsoft's cloud-based threat intelligence to help identify emerging threats.
- **Automatic sample submission**: Can send suspicious samples to Microsoft for further analysis, depending on configuration.
- **Tamper Protection**: Helps prevent unauthorized changes to important security settings.

Scan options: Microsoft Defender provides different scanning methods:

Scan type	Purpose
Quick scan	Checks common locations where threats may be found
Full scan	Scans files and running programs across the system
Custom scan	Scans selected files or folders
Microsoft Defender Offline scan	Scans outside the normal Windows environment

To inspect Defender status using PowerShell:

```
Get-MpComputerStatus
```

To inspect configured preferences:

```
Get-MpPreference
```

Practical exercise: Explore Virus & threat protection and review the available scan options. Avoid disabling protection or running scans against files you do not own or have permission to inspect.

Task 5 - Firewall & Network Protection

Windows Defender Firewall is a host-based firewall that controls network traffic entering and leaving a Windows computer.

A firewall helps protect a device by applying rules to network connections.

Windows commonly provides different network profiles:

Profile	Typical environment
Domain network	Organization-managed domain environment
Private network	Trusted home or work network
Public network	Untrusted networks such as public Wi-Fi

The firewall can apply different rules depending on the active network profile.

Inbound and outbound traffic

- Inbound traffic: Connections coming into the computer.
- Outbound traffic: Connections leaving the computer.

Firewall rules can allow or block traffic based on factors such as application, port, protocol, and network profile.

Useful PowerShell commands:

```
Get-NetFirewallProfile
```

To inspect firewall rules:

```
Get-NetFirewallRule
```

To open the advanced firewall management interface:

wf.msc

Practical exercise: Explore the firewall settings and inspect the profiles available on the virtual machine. Do not disable firewall protection.

Task 6 - App & Browser Control

App & Browser Control provides protection against potentially unsafe applications, websites, and downloads.

One important component is Microsoft Defender SmartScreen.

SmartScreen helps identify potentially dangerous content, including:

- Suspicious websites.
- Phishing pages.
- Malicious downloads.
- Untrusted applications.

It can display warnings when a user attempts to access content that Windows considers potentially unsafe.

Why this matters

Attackers often use phishing websites and malicious downloads to compromise computers.

Browser and application reputation checks can help users avoid these threats.

Practical exercise: Explore App & Browser Control and identify the protection options available in the Windows Security interface.

Task 7 - Device Security

Device Security provides information about hardware-based security capabilities.

Depending on hardware and Windows edition, features may include:

- **Trusted Platform Module (TPM):** A hardware-based security component used to protect cryptographic keys and support security features.
- **Secure Boot:** Helps prevent unauthorized boot software from loading during startup.
- **Core isolation:** Uses virtualization-based security to help protect important parts of the operating system.
- **Memory integrity:** Helps protect the Windows kernel by restricting certain types of potentially malicious code execution.

Why hardware security matters

Software-based protection alone may not be enough to defend against every attack.

Hardware-backed security features provide additional protection for sensitive operations, startup processes, and encryption keys.

Practical exercise: Open Device Security and inspect the available hardware security information. Some options may not appear if the virtual machine does not support the required hardware features.

Task 8 - BitLocker

BitLocker is a Windows drive encryption feature.

It helps protect stored information by encrypting an entire drive.

If a device is lost or stolen, encryption can make it significantly harder for someone to access data stored on the drive without the required authentication.

Important BitLocker concepts

- Drive encryption: Protects stored data.
- Recovery key: Provides an alternative way to unlock an encrypted drive.
- TPM: Can securely store or protect encryption-related material.
- Startup authentication: May require additional authentication before Windows starts.

BitLocker is particularly useful for laptops and portable computers that may be lost or stolen.

To inspect BitLocker status from an elevated Command Prompt:

```
manage-bde -status
```

To open the BitLocker management interface, search for:

```
Manage BitLocker
```

Security note: Recovery keys must be stored securely. Losing access to the recovery key can make encrypted data difficult or impossible to recover.

Practical exercise: Explore the BitLocker interface and understand the information it displays. Do not enable, disable, or modify encryption on the lab machine unless the room explicitly instructs you to do so.

Task 9 - Volume Shadow Copy

Volume Shadow Copy Service (VSS) is a Windows technology that creates point-in-time copies of files or volumes.
These copies are commonly called shadow copies.

They can support:

- File recovery.
- Backup applications.
- System restoration.
- Recovery of earlier file versions.

Previous Versions
Windows can expose earlier versions of files and folders when suitable shadow copies or backups exist.

To explore this feature:

a. Open File Explorer.
b. Navigate to a file or folder.
c. Open its Properties.
d. Look for the Previous Versions tab, if available.

The tab may not contain any versions if no suitable snapshots or backups exist.

Why VSS matters in cybersecurity
Volume Shadow Copy has both defensive and security implications.

- Administrators can use it to recover previous file versions.
- Incident responders may examine shadow copies during investigations.
- Ransomware operators sometimes attempt to delete shadow copies to make recovery harder.

Useful commands include:

```
vssadmin list shadows
```

To inspect shadow storage:

```
vssadmin list shadowstorage
```

These commands are useful for understanding whether shadow copies exist and how storage is configured.

Practical exercise: Inspect the shadow-copy information available on the virtual machine. Do not delete existing snapshots.

5. Practical Exercises

Use the Windows virtual machine to explore the following features independently.

- Open Windows Update and inspect its configuration.
- Explore the Windows Security dashboard.

- Review Virus & threat protection.
- Identify the available Microsoft Defender scan types.
- Inspect firewall network profiles.
- Explore App & Browser Control.
- Review Device Security.
- Understand TPM and Secure Boot.
- Explore the BitLocker management interface.
- Learn how Volume Shadow Copy works.
- Inspect available shadow copies using vssadmin.

6. Command Cheat Sheet

Command	Description
ms-settings:windowsupdate	Open Windows Update
Get-MpComputerStatus	Inspect Defender status
Get-MpPreference	Inspect Defender preferences
Get-NetFirewallProfile	Display firewall profiles
Get-NetFirewallRule	List firewall rules
wf.msc	Open advanced firewall settings
manage-bde -status	Display BitLocker status
vssadmin list shadows	List shadow copies
vssadmin list shadowstorage	Display shadow storage configuration

7. Key Takeaways

After studying this room, you should understand:

a. Windows Update delivers security patches and operating system improvements.
b. Windows Security provides a central interface for built-in protection features.
c. Microsoft Defender Antivirus helps detect and prevent malware.
d. Windows Defender Firewall controls network traffic.
e. SmartScreen helps protect against suspicious applications and websites.
f. TPM and Secure Boot provide hardware-backed security capabilities.
g. BitLocker protects stored data through drive encryption.
h. Volume Shadow Copy supports recovery of earlier file versions.
i. Shadow copies can be valuable during incident response and recovery.

8. Conclusion

Windows Fundamentals 2 introduces the security and recovery mechanisms built into Windows.

These features are important for protecting systems against malware, network attacks, unauthorized access, and data loss.

For cybersecurity professionals, understanding these tools is useful when auditing Windows configurations, investigating compromised systems, and developing defensive security practices.